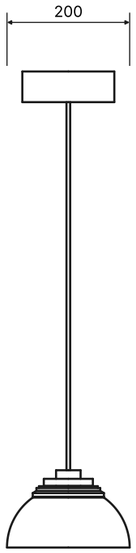
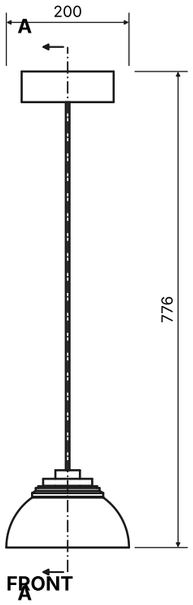
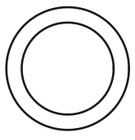
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

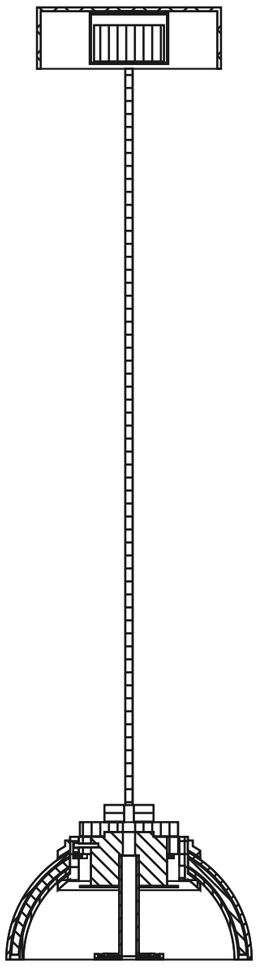
General arrangement

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Section A-A (sheet 1): the product cut at x 0.0 and seen from the right; cut faces hatched, a different angle for each part.



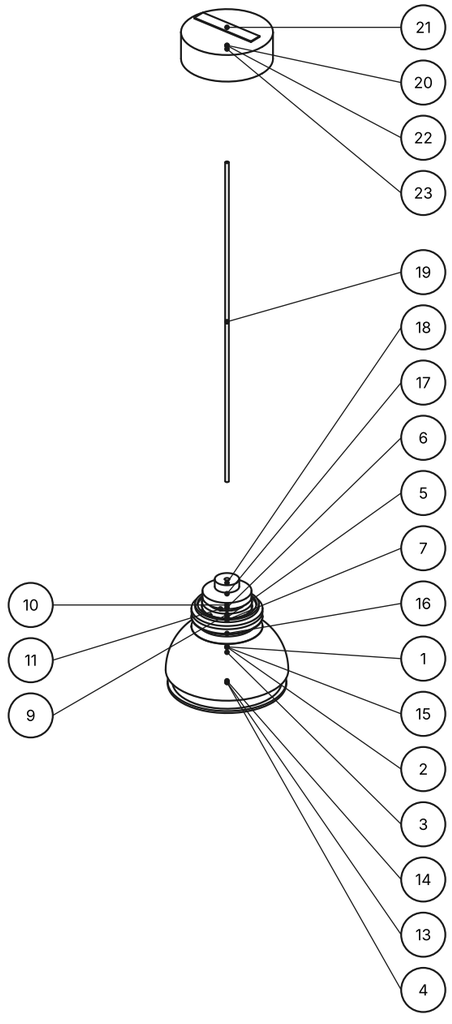
Section A-A

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

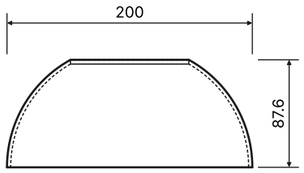
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

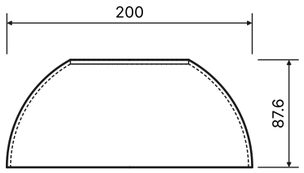
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. shell: formed, acrylic-formed; 160 cm3 each; 1 for the set; 1:5

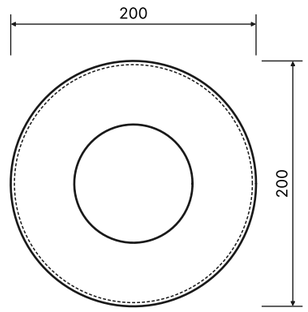
v3, the outside: 1/8 in (3 mm) stock TruCast in the line's code (Ember 2423, Poison 2092, Haze 2287, Dust 2422, Acid 9097, Ice 2069, Fog 0000, Smoke 2074), drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds; the one tinted layer, seen unlit, glowing lit (15 to 20 % at 3 mm for 2423 and 2092). Its mould is v2's shell mould (the outer face, R 100), the sheet 1/8 in instead of 10 mm



FRONT



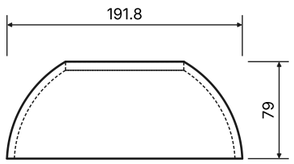
FROM THE RIGHT



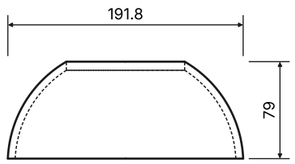
FROM ABOVE

2. liner: formed, acrylic-clear-formed; 265 cm3 each; 1 for the set; 1:5

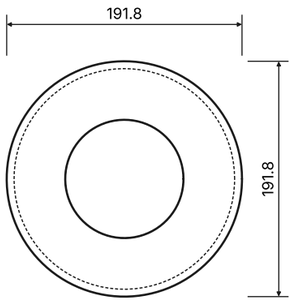
v3, the middle: 6 mm clear cast sheet (TruCast 0000), drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds: the structural shell (R 96 to 90), clamped at the crown with the others; a new mould (v2's liner mould was R 90)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

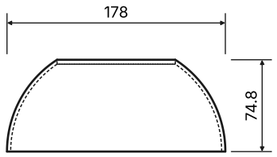
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

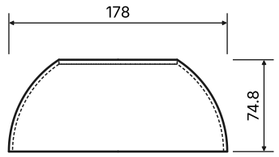
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. skin: formed, acrylic-opal-formed; 120 cm3 each; 1 for the set; 1:5

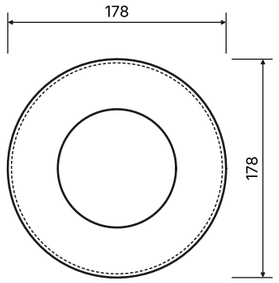
v3, the inside: 3 mm volume-diffusing (opal) cast acrylic, Perspex GS 041 or equal (T 54 %, R 39 %, ASTM D1003), drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds, formed into a female mould (R 89 to 86, a new mould), so its seen face keeps its gloss; it hides the COB and lights evenly



FRONT



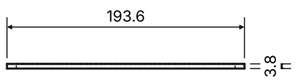
FROM THE RIGHT



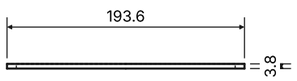
FROM ABOVE

4. rim: lathe, cast-acrylic; 17 cm3 each; 1 for the set; 1:5

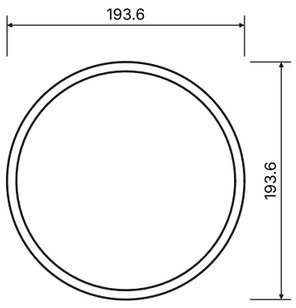
closes the three layers at the mouth, bonded and polished with them; a press-in ePTFE vent in its face keeps the gaps between the layers at room pressure and dry. PROPOSAL 2026-11-03 (the night pictures: a lit polished edge at the mouth pipes the diffuser's light out as a bright blue line): v3's opal ends 4 inside the rim, as the clear shell does, and the rim's exposed face is bead-blasted satin, not polished: no lit polished edge faces the room; about +3*to* + 5 a shade for the blasting [E], the opal's shorter trim free



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

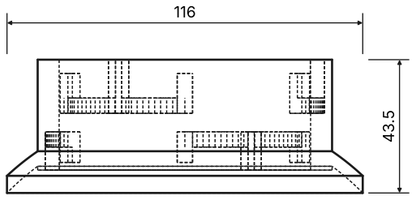
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

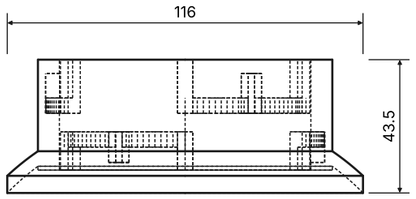
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. collar: lathe, aluminium; 82 cm3 each; 1 for the set; 1:2

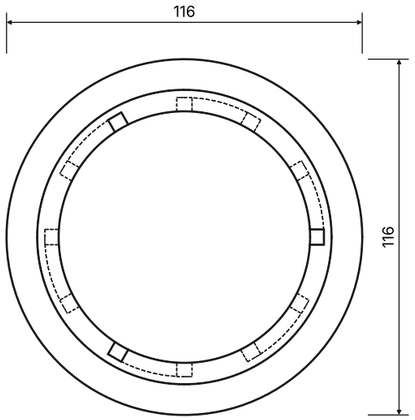
aluminium 6061, black anodised, satin: a tube through the crown with a spherical flange under the skin; three blind L-slots with T-ended notches (the quarter-turn bayonet, 30 deg) work either way up; M96 × 1.5 outside for the clamp ring. It carries the cover mechanically: the bond is a seal only



FRONT



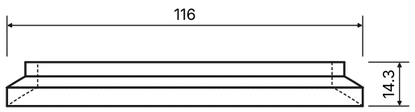
FROM THE RIGHT



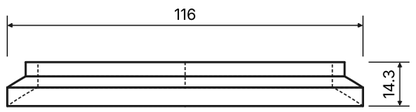
FROM ABOVE

6. clamp: lathe, aluminium; 24 cm3 each; 1 for the set; 1:2

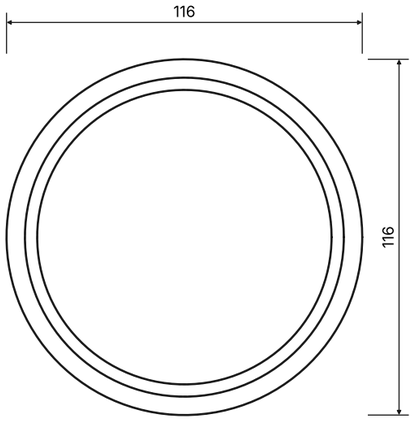
aluminium, black anodised: screwed onto the collar over the crown, its spherical face on a silicone ring, to a torque the shade maker sets (the acrylic is clamped between two silicone rings, never metal on acrylic); no thread-locker near acrylic (review 2026-11-01: anaerobics craze it): held by the silicone's preload and a nylon-tipped set screw through the ring into the collar



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

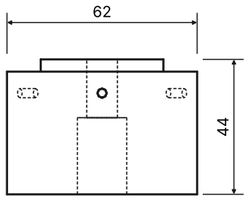
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

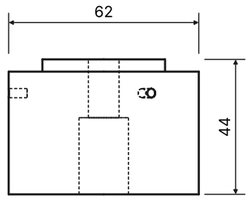
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

9. hub: lathe, aluminium; 119 cm3 each; 1 for the set; 1:2

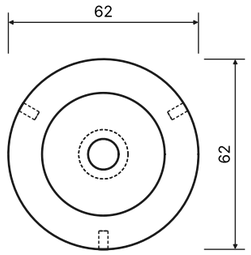
aluminium 6061, black anodised: the core, turned from bar. The COB on its holder at the top of the 46 bore (22 up: the cut-off, about 46 deg from the vertical), the reflector in the bore, the PEEK ring round it, the crown on its neck on a thermal pad; a 10 bore for the wires



FRONT



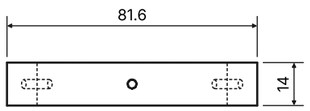
FROM THE RIGHT



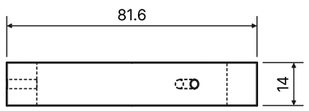
FROM ABOVE

10. ring: lathe, peek; 31 cm3 each; 1 for the set; 1:2

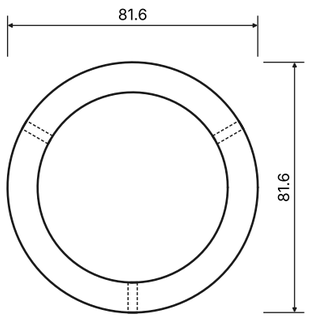
PEEK, unfilled (0.25 W/m K), black: the thermal break. It carries the three pins and the detent, so the cover touches the fixture only through it. Review 2026-11-01: its 81.6 OD is the centering land in the collar's 82 bore (0.2 diametral), three Ø3.2 clearance holes for the pins' threads



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

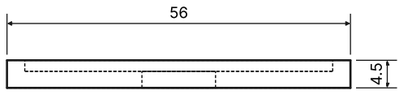
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

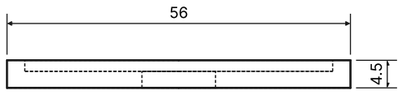
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

13. carrier: lathe, aluminium; 7 cm3 each; 1 for the set; 1:1

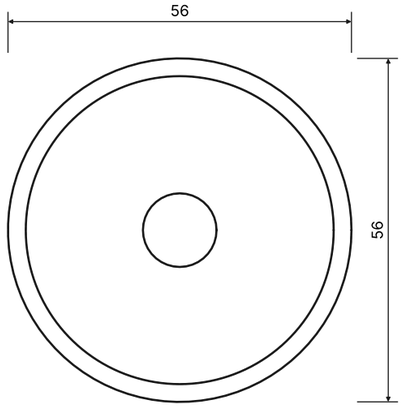
option 3 (2026-11-10): aluminium 6061, black anodised, Ø56 × 4.5, its underside on the rim plane; the board in a Ø50.2 pocket 1.8 deep in its top, so the LEDs' faces sit 4.9 above the rim plane



FRONT



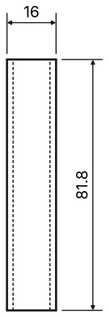
FROM THE RIGHT



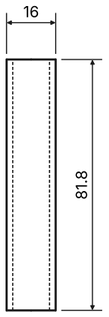
FROM ABOVE

15. stem: tube, aluminium; 7 cm3 each; 1 for the set; 1:2

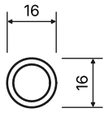
16/12 aluminium tube, black anodised, from the carrier up into the hub's socket, bonded and cross-pinned; the wires up its bore; a heat pipe inside it optional (about 2.5 W of heat)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

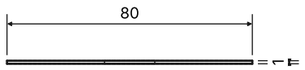
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

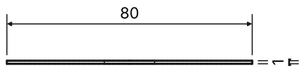
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

16. white-disc: laser, aluminium; 5 cm3 each; 1 for the set; 1:2

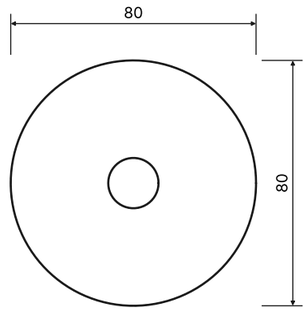
a white Ø80 × 1 disc under the hub (white powder coat), and the collar's flange face inside painted white: the crown's inside white, so the light the shade sends back up returns down



FRONT



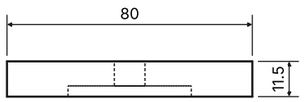
FROM THE RIGHT



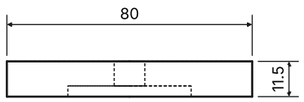
FROM ABOVE

17. crown: lathe, aluminium; 53 cm3 each; 1 for the set; 1:2

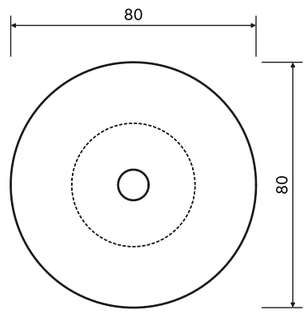
the heat sink (round 41; the fins until then), round 32f's low top: 80 across, 11.5 tall over the collar, turned from 6061 bar, black anodised (emissivity about 0.85: half its loss is radiation), on the core's neck on a thermal pad, 0.5 clear of the PEEK ring and the collar



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

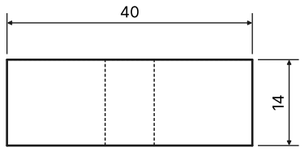
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

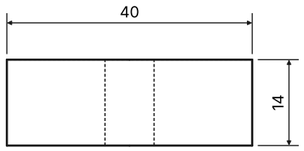
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

18. gland: lathe, aluminium; 17 cm3 each; 1 for the set; 1:1

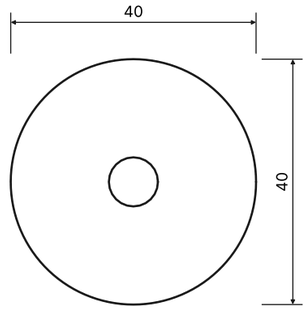
the cap, 40 × 14: aluminium, black anodised and polished (round 32f's was black acrylic: here it runs at the hub's 49 to 52 C), over the listed cord grip (the strain relief, tested with the fixture's weight), on two M3 into the crown; part of the heat sink



FRONT



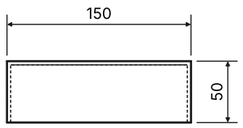
FROM THE RIGHT



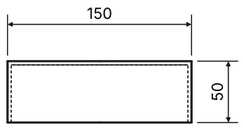
FROM ABOVE

20. canopy: lathe, cast-acrylic; 118 cm3 each; 1 for the set; 1:5

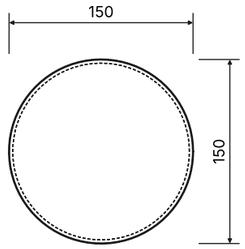
150 × 50, black cast acrylic, polished: a cover only (review 2026-11-01), screwed to the steel strap, which carries the load; the driver and the splices are in the steel enclosure



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	shell	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	160 cm3, 210 × 210 × 98	1,515.53
2	liner	formed	1	Cast acrylic sheet, clear (TruCast 0000), drape-formed over an aluminium mould	265 cm3, 202 × 202 × 89	1,519.57
3	skin	formed	1	Perspex GS 041 3 mm opal, or equal (T 54%, R 39%, ASTM D1003): cast acrylic sheet, volume-diffusing, formed over an aluminium mould	120 cm3, 188 × 188 × 85	1,212.95
4	rim	lathe	1	Cast acrylic tube, tinted through	204 × 204 × 14	67.88
5	collar	lathe	1	Aluminium 6061	126 × 126 × 53	66.22
6	clamp	lathe	1	Aluminium 6061	126 × 126 × 24	54.65
7	gaskets	buy	1	Two moulded silicone rings (Shore 40, black, 1 mm, spherical): one between the collar's flange and the skin inside, one between the clamp ring and the shell outside		3.00
8	vent	buy	1	Gore or alike A press-in ePTFE pressure-equalising vent (about 4 across), in the rim ring: it lets the air gaps between the layers breathe and keeps out dust and liquid		1.50
9	hub	lathe	1	Aluminium 6061	72 × 72 × 54	52.00
10	ring	lathe	1	PEEK rod, unfilled	92 × 92 × 24	115.48
11	pins	buy	1	Three stainless shouldered pins, M3 thread with a Ø4 × 10.5 shoulder, screwed through the PEEK ring into the hub, a pre-applied nylon patch (no anaerobic locker: it crazes acrylic); 4 mm or more of bayonet engagement against a centering		6.00
12	tether	buy	1	Review 2026-11-01: the shade's secondary retention, a 1 mm stainless wire-rope tether (7×7), crimped loops, from an eye screwed into the hub to an eye on the collar, slack in use; rated well over 10 x the 1.1 kg shade		4.00
13	carrier	lathe	1	Aluminium 6061	66 × 66 × 14	46.58
14	board	buy	1	Samsung (or Lumileds / Nichia equivalent, to choose) Hemisphere option 3 (the owner, 2026-11-10: "Core cut in half"): a flat Ø50 aluminium-core ring board, 24 mid-power 3030 LEDs (Samsung LM301H EVO class, CRI 90, 2700 K; 3000 K for Ice and		14.00
15	stem	tube	1	Aluminium 6061	26 × 26 × 92	46.55
16	white-disc	laser	1	Aluminium 6061	90 × 90 × 11	47.23
17	crown	lathe	1	Aluminium 6061	90 × 90 × 22	49.35
18	gland	lathe	1	Aluminium 6061	50 × 50 × 24	46.50
19	cord	buy	1	any (UL listed) Black textile-braided pendant cord, 2-conductor 18 AWG, UL listed (SVT-rated, or a listed Class 2 cord: past the driver in the canopy it carries the LED's 26 V DC, not mains), 3 m cut to the drop; a listed cord grip in the cap and		16.00
20	canopy	lathe	1	Cast acrylic tube, tinted through	160 × 160 × 60	106.44
21	strap	buy	1	any (a lighting supplier) Zinc-plated steel crossbar, 1.6 mm, screwed to the 4 in junction box; the cord grip, the suspension gripper and the canopy's screws anchor in it, and the driver and the 120 V splices sit in a closed steel enclosure on it.		6.00
22	enclosure	buy	1	to make (a sheet-metal shop) or a listed driver enclosure Review 2026-11-01: a closed steel enclosure, 92 × 64 × 37, 1.0 wall, zinc-plated, round the driver and the 120 V splices, screwed to the canopy strap; a listed bushing where the Class 2		12.00

Parts list (1 of 2)

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
23	driver	buy	1	ERP Power ESS010W-0180-42 (the 1,000 lm level, 2026-11-10; the same driver): constant current 180 mA, 7.6 W, 24-42 V DC out, 120-277 V AC in; forward-phase (TRIAC) and reverse-phase (ELV) dimming at 120 V AC and 0-10 V, 1-100 %; Class 2;		16.00
24	plunger	buy	1	Kipp or Misumi A stainless ball spring plunger M5, light force (about 10 N), threaded into the hub: the bayonet's detent, clicking into a dimple in the collar		4.50
25	listing	buy	1	UL or Intertek (ETL) Listing the pendant to UL 1598 (and the table lamp's battery and charger to UL 153 / IEC 62368): a one-off cost a design, spread over the first run		[PRICE]
26	suspension	buy	1	[option: suspension] Griplock Systems or Gripple (to choose) Round 43, PROPOSAL (certification.md): a 1.5 mm black-coated stainless suspension cable, 2 m, cut to the drop, with an adjustable gripper in the canopy and a fixed end in the cap;		12.00

Parts list (2 of 2)

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. Option 3 (the owner, 2026-11-10: "it should work like Core cut in half"; the main build): the light sits at the mouth's centre and faces up into the shade, so the shade glows in its tint and the light cast down is the bulb's. A Ø50 ring board of 24 mid-power LEDs (two strings of 12 at about 90 mA, about 1,000 lm, about 200 lux under the lamp) on a black Ø56 × 4.5 carrier, its underside on the rim plane, the LEDs' faces about 5 above it; a 16/12 black anodised stem up to the hub; the crown's inside white (a white Ø80 disc under the hub, the collar's flange face white); the opal to the rim. The COB, its reflector and holder are gone; the ERP driver, the Class 2 cord, the canopy and the UL 1598 path stay. Without a lens until a stock side-emitting ring optic with an IES file is named. Heat test needed: the hub about 6 K warmer than today (estimate). The notes below describe the COB build.
2. Hemisphere (the Dome until round 35; Hemisphere North in rounds 35 to 37): the shade hangs mouth down. Its shades swap by hand: lift the shade 5 mm, turn it 30 degrees and lower it off. To fit one, line up the three slots, push up, turn until the plunger clicks, and let go: its weight holds each pin up in its notch, so a knock cannot turn it off. No tools, and the shade is never live (the COB, the hub and the driver stay up).
3. The shade is clamped, not hung on a glue joint: the collar's flange sits under the crown inside, a ring screws down over it outside, the acrylic between two silicone rings. A failed bond cannot drop it.
4. The wall is tint, opal, tint: a 9 mm tinted shell, a 3 mm volume-diffusing opal liner, a 3 mm tinted skin, each formed from cast sheet on an aluminium mould, nested, closed at the mouth by a tinted rim with a vent. Opal scatters through its depth, so it still diffuses where it touches the tint (a bead-blasted clear liner would go clear wherever it was bonded). Nothing white is seen.
5. Light: a Bridgelux Vero SE 10 (2700 K, CRI 90) at the top of a 46 bore lined with a reflector, 22 up: the cut-off about 46 deg from the vertical, and about 85 % of its light out (the black bore passed 52 %). About 575 lm at the COB (the owner, round 36): about 590 at 180 mA (4.4 W) on an ERP Power ESS010W-0180-42 (Class 2, FCC Part 15 Class B), dimmed 100 % to 1 % by a phase-cut (TRIAC or ELV) dimmer or 0-10 V. Not for 1,100 lm: the low top is sized for 575 (1,100 needs the fins of rounds 35 to 39).
6. Heat (round 41): round 32f's low top is the heat sink, no fins: a black anodised aluminium crown 80 across and 11.5 tall over the collar, and an aluminium cap 40 × 14 over the cord grip, the cap's top 126 above the rim. A PEEK ring carries the bayonet's pins, so the hub reaches the shade only through it and air. At 575 lm the hub makes 2.7 W of heat; estimated in a 35 C room: the hub 49 C, the COB's case 50 C, the collar and the acrylic beside it 41 C (hemisphere_thermal.py; to measure, README round 35).
7. Mains stops at the canopy: the driver is there, and the cord carries the LED's 26 V DC (Class 2). The cord is a listed 18 AWG 2-conductor in a black textile braid, its grips tested with the fixture's weight. To be listed to UL 1598 (hardwired) with its own shades, each tint in the listing; the canopy's strap and steel enclosure carry the grip, the gripper and the splices, the acrylic a cover only.
8. Hang over a bar at 750 to 900 centres, the rim 700 to 800 above the counter; over a table, one a table, the rim about 650 to 750 above it.

Notes

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

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Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15